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Convert Local Coordinate Systems to Standard Coordinate Systems

Using 2D Conformal Transformation in MicroStation V8i and Bentley Map V8i

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How to compute and apply rotation, scaling and XY translation to convert local coordinate systems to standard coordinate systems.

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Introduction

A 2D conformal transformation is used to transform from one rectangular coordinate system to another rectangular coordinate system when the two coordinate systems differ from each other by up to four parameters: scale, rotation, translation in the X direction, and translation in the Y direction. When these four transformation parameters are known, any point whose XY coordinates are known in one system can be transformed to the other system. This transformation is equivalent to a combined Rotate, Scale, and Move or Fence Rotate, Fence Scale and Fence Move operation in MicroStation. The Rotate and Scale must be done relative to the origin of the coordinate system being transformed ("xy=0,0"), and the Move must be the last of the three operations.

When the four parameters are unknown, two horizontal control points whose coordinates are known in both coordinate systems can be used to compute the transformation parameters. The coordinates of the points can be determined from a CAD drawing or from calculated coordinates of control points from survey observations. When there are more than two points, a least squares solution for the transformation parameters can be computed.

A 2D conformal transformation is not intended to be used to transform from a standard coordinate system as adopted by a governmental agency for a defined region, such as a State Plane Coordinate System zone or a zone in the UTM coordinate system, to another standard coordinate system. A reprojection is required in that case. A reprojection can be performed in MicroStation V8i simply by changing the coordinate system of the design file from one coordinate system to another. A 2D conformal transformation should not be ruled out completely though. For small areas it may be acceptable and beneficial. For example, orthogonal grid or index lines in one standard coordinate system will not remain straight or orthogonal when reprojected to another coordinate system. In a 2D conformal transformation, straight lines will remain straight and orthogonal lines will remain orthogonal.

The terms "local" and "standard" are used to demonstrate the process of performing a transformation in regard to surveying and mapping. However the same process applies to any transformation of any rectangular coordinate system.

In this document unknown variables that are to be computed are displayed as characters with an italic font style. Matrices are uppercase bold characters. A bold italic font is used to indicate that the matrix is to be computed. The same equation can be used in two or more sections with differing font styles depending upon which variables are known or unknown in the discussion.

Common Reasons for Performing a 2D Conformal Transformation in Surveying and Mapping

In surveying and mapping there are three common reasons to perform a 2D conformal transformation.

- An arbitrary or convenient local coordinate system has been established by survey and needs to be converted to a standard coordinate system.
- An arbitrary or convenient coordinate system has been used in CAD drawings and needs to be converted to a standard coordinate system.
- A civil engineering project coordinate system at ground level needs to be referenced to a standard coordinate system. A common misconception is that a perfect ground survey will exactly match a perfect map using a standard map projection and coordinate system, and therefore will have a scale factor of 1.0 relative to the map. This is not true. Grid to ground scale factors are discussed in Compute the Grid to Ground Scale Factor as an Affine Parameter.

Mathematically Transform Points from a Local Coordinate System to a Standard Coordinate System

Mathematically convert points in a local rectangular coordinate system to points in a standard coordinate system such as the UTM or U.S. State Plane Coordinate System, where the four parameters are known. The conversion for a single point is as shown below.

Transformation Equations for a Point

$$E = k\,X\,\text{Cos}\,\theta - k\,Y\,\text{Sin}\,\theta + t_x$$

$$N = k\,X\,\text{Sin}\,\theta + k\,Y\,\text{Cos}\,\theta + t_y$$

Where the parameters are:

k = the ratio of the distance between 2 points in a standard coordinate system to the distance between the same 2 points in a local coordinate system

θ = rotation about the origin of the local coordinate system to make it parallel with the standard system (counter-clockwise is positive)

t_x = translation in the X direction applied in the local system after the scale and rotation are applied

t_y = translation in the Y direction applied in the local system after the scale and rotation are applied

And:

X and Y are the known coordinates of a point in the local coordinate system.

E and N are the unknown easting and northing of a point in a standard coordinate system.

Or

$$E = a\,X - b\,Y + t_x$$

$$N = a\,Y + b\,X + t_y$$

Where

$$a = k\,\text{Cos}\,\theta$$

$$b = k\,\text{Sin}\,\theta$$

To transform numerous points from one coordinate system to the other, matrix algebra is more convenient. See the Appendix for instructions on using Microsoft Excel to perform matrix calculations. Note the data entry pattern for the **X** coefficient matrix. A single point requires 2 rows in the **X** coefficient matrix and produces the transformed coordinates in the corresponding 2 rows of the ***E*** column matrix.

$$\boldsymbol{E} = \mathbf{X}\,\mathbf{T}$$

Where:

$$\boldsymbol{E} = \begin{vmatrix} E_1 \\ N_1 \\ E_2 \\ N_2 \\ . \\ . \\ E_n \\ N_n \end{vmatrix} \qquad \mathbf{X} = \begin{vmatrix} X_1 & -Y_1 & 1 & 0 \\ Y_1 & X_1 & 0 & 1 \\ X_2 & -Y_2 & 1 & 0 \\ Y_2 & X_2 & 0 & 1 \\ . & . & . & . \\ . & . & . & . \\ X_n & -Y_n & 1 & 0 \\ Y_n & X_n & 0 & 1 \end{vmatrix} \qquad \mathbf{T} = \begin{vmatrix} k\,\mathrm{Cos}\,\theta \\ k\,\mathrm{Sin}\,\theta \\ t_x \\ t_y \end{vmatrix}$$

Example 1 – Mathematically Transform Points with Known Transformation Parameters

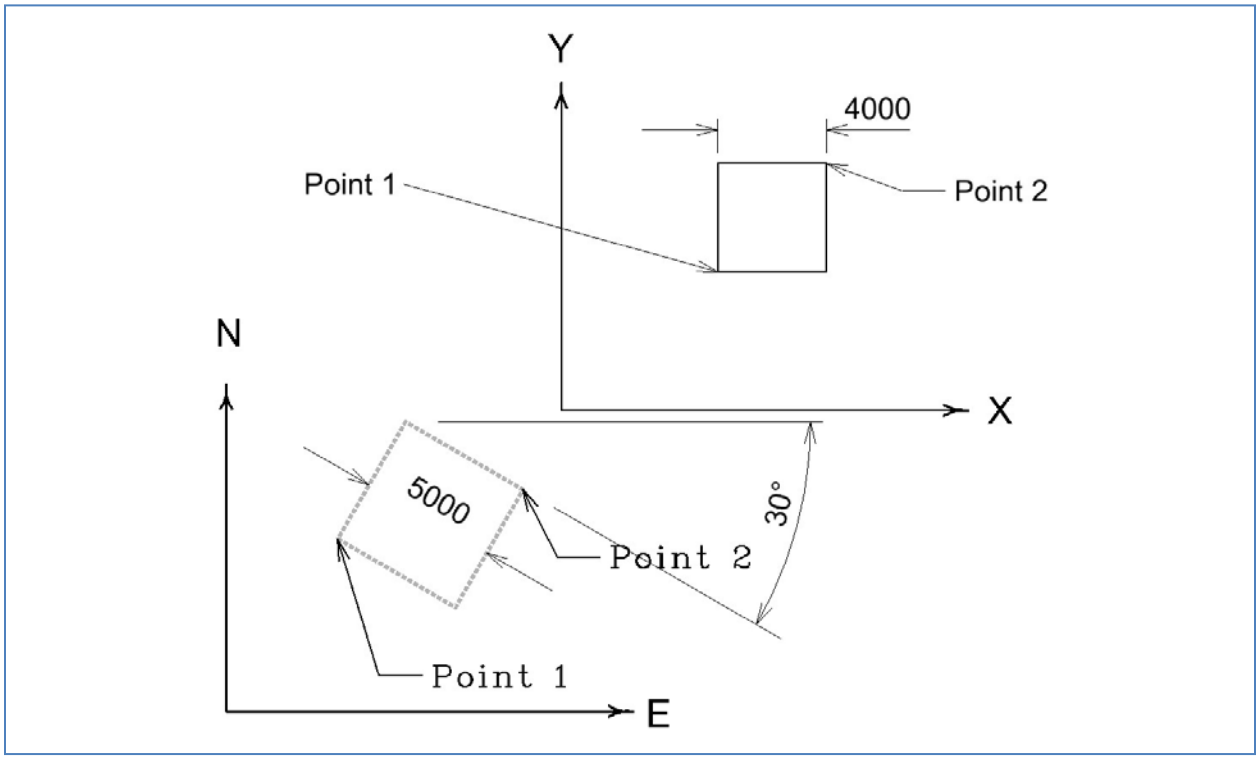


Figure 1

Transform the corners of the 4000 x 4000 square in the **XY** coordinate system to the corners of the 5000 x 5000 square in the **EN** coordinate system. The rotation is shown. The scale can be computed (5000/4000). Normally two pairs of control points in each coordinate system **XY** and **EN** are required to compute transformation parameters. With the scale and rotation known, only one pair of control points is needed to compute t_x and t_y.

Point 1 in the **EN** coordinate system

$E_1 = 397871.235$

$N_1 = 1537022.683$

Point 1 in the **XY** coordinate system

$X_1 = 100000$

$Y_1 = 200000$

Modify the transformation equations:

$$E = a\,X - b\,Y + t_x$$

$$N = b\,X + a\,Y + t_y$$

$$t_X = E - a\,X + b\,Y$$

$$t_Y = N - a\,Y - b\,X$$

Where

$$a = k \cos \theta$$

$$b = k \sin \theta$$

Compute the transformation parameters a, b, t_X and t_Y:

$\theta = -30°$

$k = 5000/4000 = 1.25$

$a = k \cos \theta$

$\boldsymbol{a} = 1.08253175473055$

$b = k \sin \theta$

$\boldsymbol{b} = -0.625$

$t_x = E - k\,X \cos \theta + k\,Y \sin \theta$

$t_x = 397871.235 - 1.25\,(100000) \cos(-30°) + 1.25\,(200000) \sin(-30°)$

$\boldsymbol{t_x} = 164618.060$

$t_y = N - k\,X \sin \theta - k\,Y \cos \theta$

$t_y = 1537022.683 - 1.25\,(100000) \sin(-30°) - 1.25\,(200000) \cos(-30°)$

$\boldsymbol{t_y} = 1383016.332$

Transform points of the four corners of the square in the **XY** coordinate system. The XY coordinates are:

Point 1 = 100000, 200000

Point 2 = 104000, 204000

Point 3 = 104000, 200000

Point 4 = 100000, 204000

Note the data entry pattern in the coefficient matrix **X**.

$$E = \mathbf{X}\,\mathbf{T}$$

$$\mathbf{X} = \begin{bmatrix} 100000 & -200000 & 1 & 0 \\ 200000 & 100000 & 0 & 1 \\ 104000 & -204000 & 1 & 0 \\ 204000 & 104000 & 0 & 1 \\ 104000 & -200000 & 1 & 0 \\ 200000 & 104000 & 0 & 1 \\ 100000 & -204000 & 1 & 0 \\ 204000 & 100000 & 0 & 1 \end{bmatrix} \qquad \mathbf{T} = \begin{bmatrix} 1.08253175473055 \\ -0.625 \\ 164618.060 \\ 1383016.332 \end{bmatrix}$$

$$E = \begin{bmatrix} 397871.235 \\ 1537022.683 \\ 404701.362 \\ 1538852.810 \\ 402201.362 \\ 1534522.683 \\ 400371.235 \\ 1541352.810 \end{bmatrix}$$

Graphically Transform MicroStation Graphics with Known Transformation Parameters

The transformation parameters can be applied to graphics using three MicroStation Fence operations. The calculations from the previous examples are used to describe the process.

1. In the design file with the local **XY** coordinate system, **Fence Rotate** the graphics about the origin of the local coordinate system as the pivot point using a rotation angle of -30°. Key in "xy=0, 0" when prompted for the *pivot point*.
2. **Fence Scale** the rotated graphics by the scale factor of 1.25 using the key-in "xy=0, 0" when prompted for the *origin*.
3. **Fence Move** the graphics by selecting any point using a data point in the MicroStation window as the *first point* and key in the translation
   "dx = t_x, t_y"

   $$\mathrm{dx} = 164618.060,\ 1383016.332$$

   as the *point to define distance and direction*.

The sequence of the Fence Rotate and Fence Scale can be reversed, but the Fence Move must be the last operation.

The local coordinate system graphics will now be in the standard coordinate system.

Compute the Four Transformation Parameters

Often the transformation parameters are unknown and must be computed. A minimum of two horizontal control points whose coordinates are known in both systems are required. The point coordinates can be computed from survey observations or determined from CAD files. The two pairs of points provide two equations each for a total of four equations to compute four unknowns.

Points **1** and **2** in the standard coordinate system **EN** correspond to Points **1** and **2** in the local coordinate system **XY**. Note that the unknown variables are italicized in the transformation equations.

$$E_1 = k X_1 \, Cos\, \theta - k Y_1 \, Sin\, \theta + t_x$$

$$N_1 = k X_1 \, Sin\, \theta + k Y_1 \, Cos\, \theta + t_y$$

$$E_2 = k X_2 \, Cos\, \theta - k Y_2 \, Sin\, \theta + t_x$$

$$N_2 = k X_2 \, Sin\, \theta + k Y_2 \, Cos\, \theta + t_y$$

To enable the use of matrix algebra to solve linear equations to obtain the transformation parameters, as before let:

$$a = k \, Cos\, \theta$$

$$b = k \, Sin\, \theta$$

And arrange the transformation parameters a and b in sequence:

$$E_1 = a X_1 - b Y_1 + t_x$$

$$N_1 = a Y_1 + b X_1 + t_y$$

$$E_2 = a X_2 - b Y_2 + t_x$$

$$N_2 = a Y_2 + b X_2 + t_y$$

Using matrix algebra:

$$\mathbf{E} = \mathbf{X}\,\boldsymbol{T}$$

Where:

$$\mathbf{E} = \begin{vmatrix} E_1 \\ N_1 \\ E_2 \\ N_2 \end{vmatrix} \qquad \mathbf{X} = \begin{vmatrix} X_1 & -Y_1 & 1 & 0 \\ Y_1 & X_1 & 0 & 1 \\ X_2 & -Y_2 & 1 & 0 \\ Y_2 & X_2 & 0 & 1 \end{vmatrix} \qquad \boldsymbol{T} = \begin{vmatrix} a \\ b \\ t_x \\ t_y \end{vmatrix}$$

Solve for ***T***:

Matrix calculator software or Microsoft Excel can be used to perform the calculations. The **X** matrix must be inverted to move it to the other side of the equation. This is a laborious task if done manually but can be easily done with matrix calculator software. The number of rows must equal the number of columns in the **X** coefficient matrix in order to invert it. In this case, **X** is a 4 x 4 (square) matrix.

$$\boldsymbol{T} = \mathbf{X}^{-1}\,\mathbf{E}$$

$$\boldsymbol{T} = \begin{array}{|c|}\hline a \\ \hline b \\ \hline t_E \\ \hline t_N \\ \hline \end{array}$$

Once the transformation parameters are computed, the rotation and scale can be determined.

Solve for θ, the rotation:

$$\theta = \mathrm{Tan}^{-1}\,[\mathrm{b} / \mathrm{a}]$$

Solve for k, the scale:

$$k = \mathrm{a} / \mathrm{Cos}\,\Theta \text{ or } k = \mathrm{b} / \mathrm{Sin}\,\Theta \text{ or } k = \sqrt{\mathrm{a}^2 + \mathrm{b}^2}$$

Example 2 – Compute Transformation Parameters

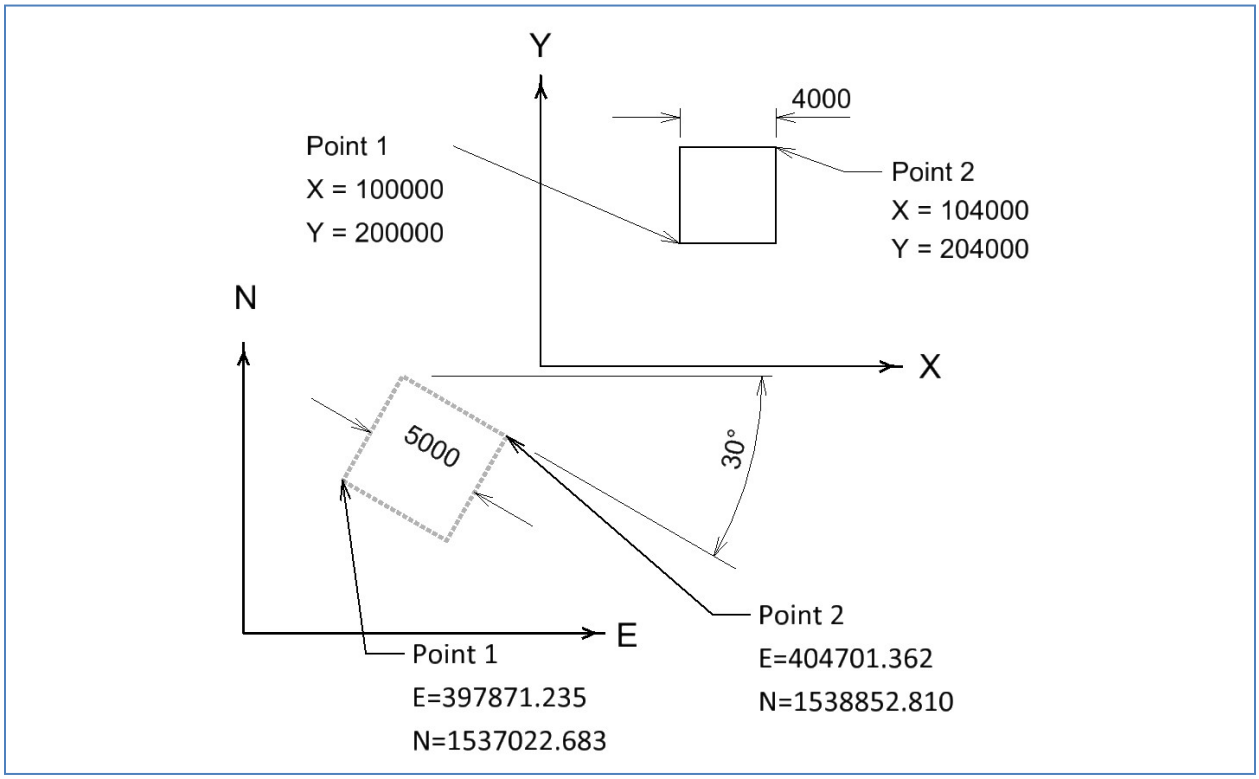


Figure 2

$$\mathbf{E} = \mathbf{X}\,\boldsymbol{T}$$

$$\mathbf{E} = \begin{vmatrix} E_1 \\ N_1 \\ E_2 \\ N_2 \end{vmatrix} \qquad \mathbf{X} = \begin{vmatrix} X_1 & -Y_1 & 1 & 0 \\ Y_1 & X_1 & 0 & 1 \\ X_2 & -Y_2 & 1 & 0 \\ Y_2 & X_2 & 0 & 1 \end{vmatrix} \qquad \boldsymbol{T} = \begin{vmatrix} a \\ b \\ t_x \\ t_y \end{vmatrix}$$

$$\mathbf{E} = \begin{vmatrix} 397871.235 \\ 1537022.683 \\ 404701.362 \\ 1538852.810 \end{vmatrix} \qquad \mathbf{X} = \begin{vmatrix} 100000 & -200000 & 1 & 0 \\ 200000 & 100000 & 0 & 1 \\ 104000 & -204000 & 1 & 0 \\ 204000 & 104000 & 0 & 1 \end{vmatrix} \qquad \boldsymbol{T} = \begin{vmatrix} a \\ b \\ t_x \\ t_y \end{vmatrix}$$

$$\boldsymbol{T} = \mathbf{X}^{-1}\,\mathbf{E}$$

$$\mathbf{X}^{-1} = \begin{vmatrix} -0.000125 & -0.000125 & 0.000125 & 0.000125 \\ 0.000125 & -0.000125 & -0.000125 & 0.000125 \\ 38.5 & -12.5 & -37.5 & 12.5 \\ 12.5 & 38.5 & -12.5 & -37.5 \end{vmatrix} \qquad \mathbf{E} = \begin{vmatrix} 397871.235 \\ 1537022.683 \\ 404701.362 \\ 1538852.810 \end{vmatrix}$$

$$\boldsymbol{T} = \begin{bmatrix} 1.08253175 \\ -0.625 \\ 164618.060 \\ 1383016.333 \end{bmatrix}$$

$\theta = \text{Tan}^{-1}$ [b / a]

$\theta = \text{Tan}^{-1}$ [−0.625 / 1.08253175]

$\theta = -30°$ (The angle is clockwise. See the previous Figure 2 above.)

$k =$ a / Cos Θ or $k =$ b / Sin Θ or $k = \sqrt{a^2 + b^2}$

$k =$ 1.08253175 / Cos (−30°)

$k =$ 1.25 (Can be checked by dividing 5000 by 4000. See the previous Figure 2.)

An exaggerated scale has been used in this example to demonstrate the process. Normally, assuming that the same linear unit of measurement is used in both coordinate systems, the scale will be exactly 1.0 or close to 1.0.

Modify the 2D conformal transformation equations to check the translation.

$$t_x = E - a\,X + b\,Y$$

$$t_y = N - b\,X - a\,Y$$

Use the coordinates of Point 1 in the standard coordinate system and the corresponding Point 1 in the local coordinate system to check the translation.

$t_x = E_1 - a\,X_1 + b\,Y_1$

$t_x = 397871.235 - 1.08253175\ (100000) - 0.625\ (200000)$

$t_x = 164618.060$ -- which equals the matrix calculation in matrix **T**

$t_y = N_1 - b\,X_1 - a\,Y_1$

$t_y = 1537022.683 + 0.625\ (100000) - 1.08253175\ (200000)$

$t_y = 1383016.333$-- which equals the matrix calculation in matrix **T**

Reference Project Coordinate Systems to Standard Coordinate Systems

Bentley Map V8i provides the capability of creating custom coordinate systems for MicroStation V8i. A civil engineering project coordinate system can be referenced to a standard coordinate system by adding a special type of custom map projection. The conformal transformation parameters must be known or computed. In previous examples the transformation was from a local **XY** coordinate system to a standard **EN** coordinate system. For this custom coordinate system, the transformation parameters must be from the standard coordinate system to the local project coordinate system. The method of computing the transformation parameters is the same as previously described except the standard coordinate values are entered into the coefficient matrix and the local coordinate values are entered into the column matrix – the reverse of what was done previously.

Another requirement is that the standard coordinate system must use either the Lambert Conformal Conic map projection technique or the Transverse Mercator technique. In the U.S. most State Plane zones use one or the other of the projection types.

The method of setting up a custom coordinate system is explained in the online help for Bentley Map under "Coordinate Systems Editing." Or search for the phrase "DTY" in the Bentley Map help. Copy the standard coordinate system to the section in the coordinate system library for custom coordinate systems. Edit the coordinate system and change the map projection to either the Lambert Conformal Conic with Affine (a 'fyn) Processor or Transverse Mercator with Affine Processor, whichever is appropriate.

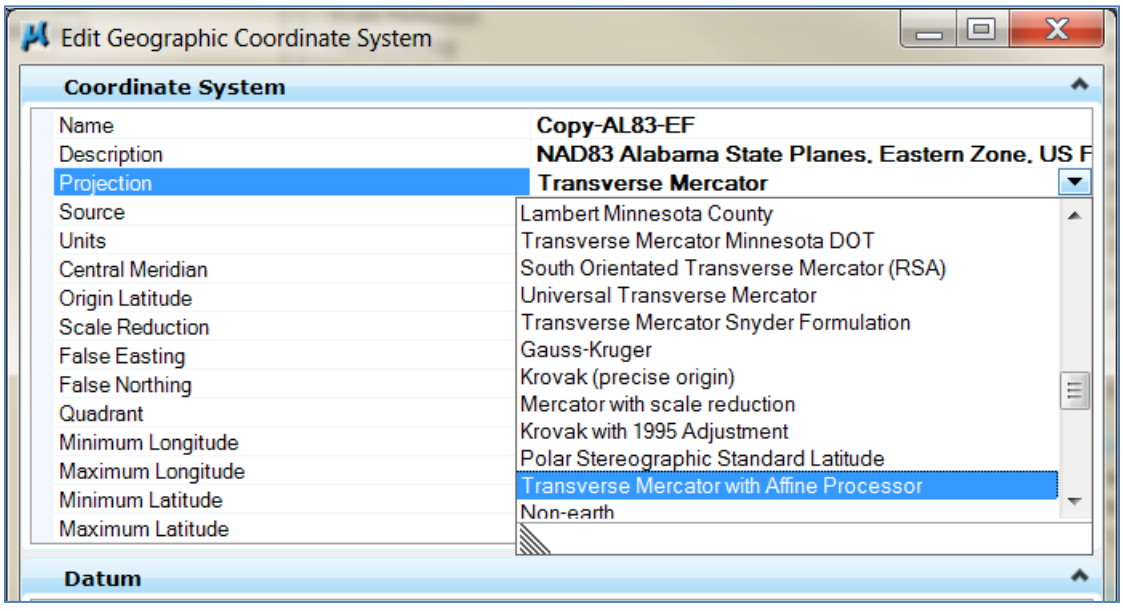


Figure 3

The affine processor parameters will be added to the coordinate system parameters.

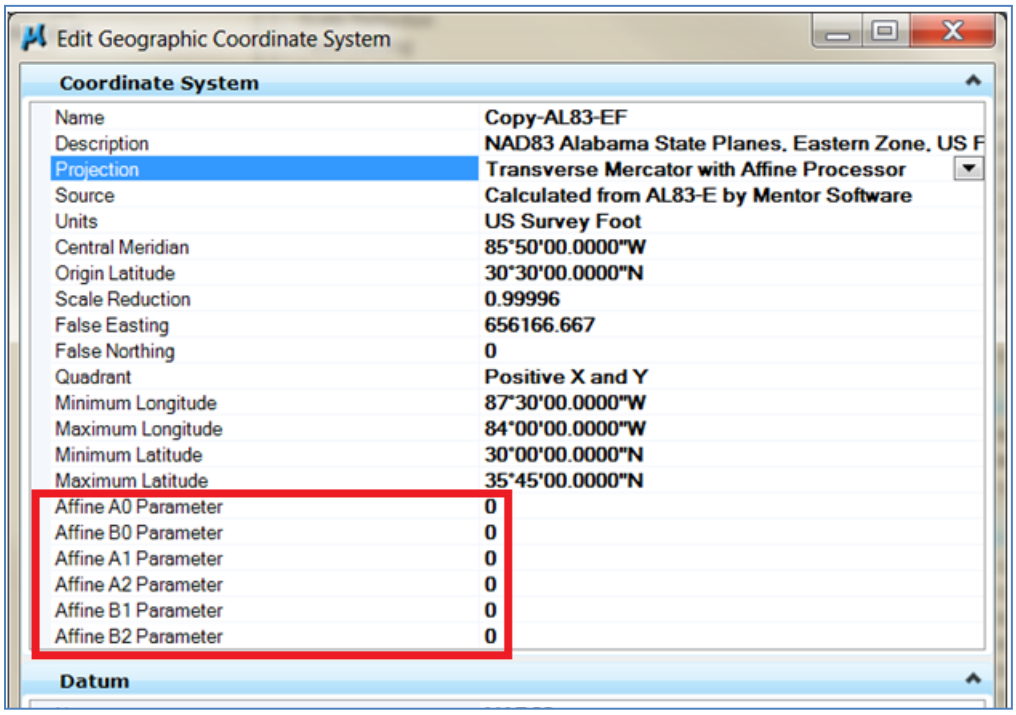


Figure 4

An affine transformation is similar to a 2D conformal transformation except that there are five parameters: rotation, the scale in the X direction, the scale in the Y direction, translation in the X direction, and translation in the Y direction. The affine transformation tools can be used to perform a 2D conformal transformation when the scale is the same in the X and Y directions.

The 2D affine equations are:

$$X = A1^{*}\,E + A2^{*}\,N + A0$$
$$Y = B1^{*}\,E + B2^{*}\,N + B0$$

The affine parameters are:

$A0 = t_E$ -- translation in the east-west direction of the standard coordinate system

$B0 = t_N$ -- translation in the north-south direction of the standard coordinate system

$A1 = k_E \cos\theta$ -- where k_E is the scale in the east-west direction and θ is the rotation

$A2 = -k_N \sin\theta$ -- where k_N is the scale in the north-south direction

$B1 = k_E \sin\theta$

$B2 = k_N \cos\theta$

Substituting in the values of the Affine Parameters:

2D Affine Transformation Equations

$$X = k_E\, E \cos\theta - k_N\, N \sin\theta + t_E$$

$$Y = k_E\, E \sin\theta + k_N\, N \cos\theta + t_N$$

When the scale is the same in the N and E direction, the affine equations become 2D conformal equations. Note that the standard coordinate system points are now being transformed to a local coordinate system. The matrix naming convention used in this document is that a matrix with standard coordinate values is called **E** while a local matrix is called **X**. In this workflow, the **E** matrix is the coefficient matrix and the **X** matrix is the column matrix.

$$\mathbf{X} = \mathbf{E}\, \boldsymbol{T}$$

$$\boldsymbol{T} = \begin{vmatrix} a \\ b \\ t_E \\ t_N \end{vmatrix} \qquad \mathbf{E} = \begin{vmatrix} E_1 & -N_1 & 1 & 0 \\ N_1 & E_1 & 0 & 1 \\ E_2 & -N_2 & 1 & 0 \\ N_2 & E_2 & 0 & 1 \end{vmatrix} \qquad \mathbf{X} = \begin{vmatrix} X_1 \\ Y_1 \\ X_2 \\ Y_2 \end{vmatrix}$$

$$\boldsymbol{T} = \mathbf{E}^{-1}\, \mathbf{X}$$

$$\boldsymbol{T} = \begin{array}{|c|} \hline a \\ \hline b \\ \hline t_E \\ \hline t_N \\ \hline \end{array}$$

$A0 = t_E$

$B0 = t_N$

$A1 = a = k \cos\theta$

$A2 = -b = -k \sin\theta$

$B1 = b = k \sin\theta$

$B2 = a = k \cos\theta$

A counter-clockwise rotation is positive.

Example 3 – Compute the 2D Affine Parameters for a Project Coordinate System

Using the same data from Example 2 but swapping coordinate point values in the column and coefficient matrices, perform a transformation from the standard coordinate to the local project coordinate system. See Figure 2.

$$\mathbf{X} = \mathbf{E}\,T$$

$$\mathbf{X} = \begin{bmatrix} 100000 \\ 200000 \\ 104000 \\ 204000 \end{bmatrix} \qquad \mathbf{E} = \begin{bmatrix} 397871.235 & -1537022.683 & 1 & 0 \\ 1537022.683 & 397871.235 & 0 & 1 \\ 404701.362 & -1538852.810 & 1 & 0 \\ 1538852.810 & 404701.362 & 0 & 1 \end{bmatrix} \qquad T = \begin{bmatrix} a \\ b \\ t_E \\ t_N \end{bmatrix}$$

$$T = \mathbf{E}^{-1}\,\mathbf{X}$$

$$T = \begin{bmatrix} 0.69282032454132 \\ 0.40000000262192 \\ 439155.799 \\ -1024029.049 \end{bmatrix}$$

$\theta = \text{Tan}^{-1}\,[\text{b} / \text{a}]$

$\theta = \text{Tan}^{-1}\,[0.4 / 0.69282032454132]$

$\theta = 30°$

Note that the sign of the angle indicates that the rotation is in the opposite direction of the calculation in Example 2. The rotation is now counter-clockwise where previously it was clockwise.

$k = \text{a} / \text{Cos}\,\Theta$ or $k = \text{b} / \text{Sin}\,\Theta$ or $k = \sqrt{\text{a}^2 + \text{b}^2}$

$k = 0.69282032454132 / \text{Cos}\,(30°)$

$k = 0.8$

Note that the scale is the inverse of the scale in Example 2 (1 / 1.25).

Modify the 2D conformal transformation equations to check the translation.

$$t_E = \text{X} - \text{a}\,\text{E} + \text{b}\,\text{N}$$

$$t_N = \text{Y} - \text{b}\,\text{E} - \text{a}\,\text{N}$$

Use the coordinates of Point 1 in the standard coordinate system and Point 1 in the local coordinate system to check the translation. Any pair of points can be used for the calculation.

$t_E = \mathrm{X}_1 - \mathrm{a}\,\mathrm{E}_1 + \mathrm{b}\,\mathrm{N}_1$

$t_E = 100000 - 0.692820324541323\ (397871.235) + 0.40000000262192\ (1537022.683)$

$t_E = 439155.799$ -- which equals the matrix calculation in matrix $\mathbf{T}$

$t_N = \mathrm{Y}_1 - \mathrm{b}\,\mathrm{E}_1 - \mathrm{a}\,\mathrm{N}_1$

$t_N = 200000 - 0.40000000262192\ (397871.235) - 0.69282032454132\ (1537022.683)$

$t_N = -1024029.049$ -- which equals the matrix calculation in matrix $\mathbf{T}$

Insert the computed transformation parameters into the custom project coordinate system Affine Parameters.

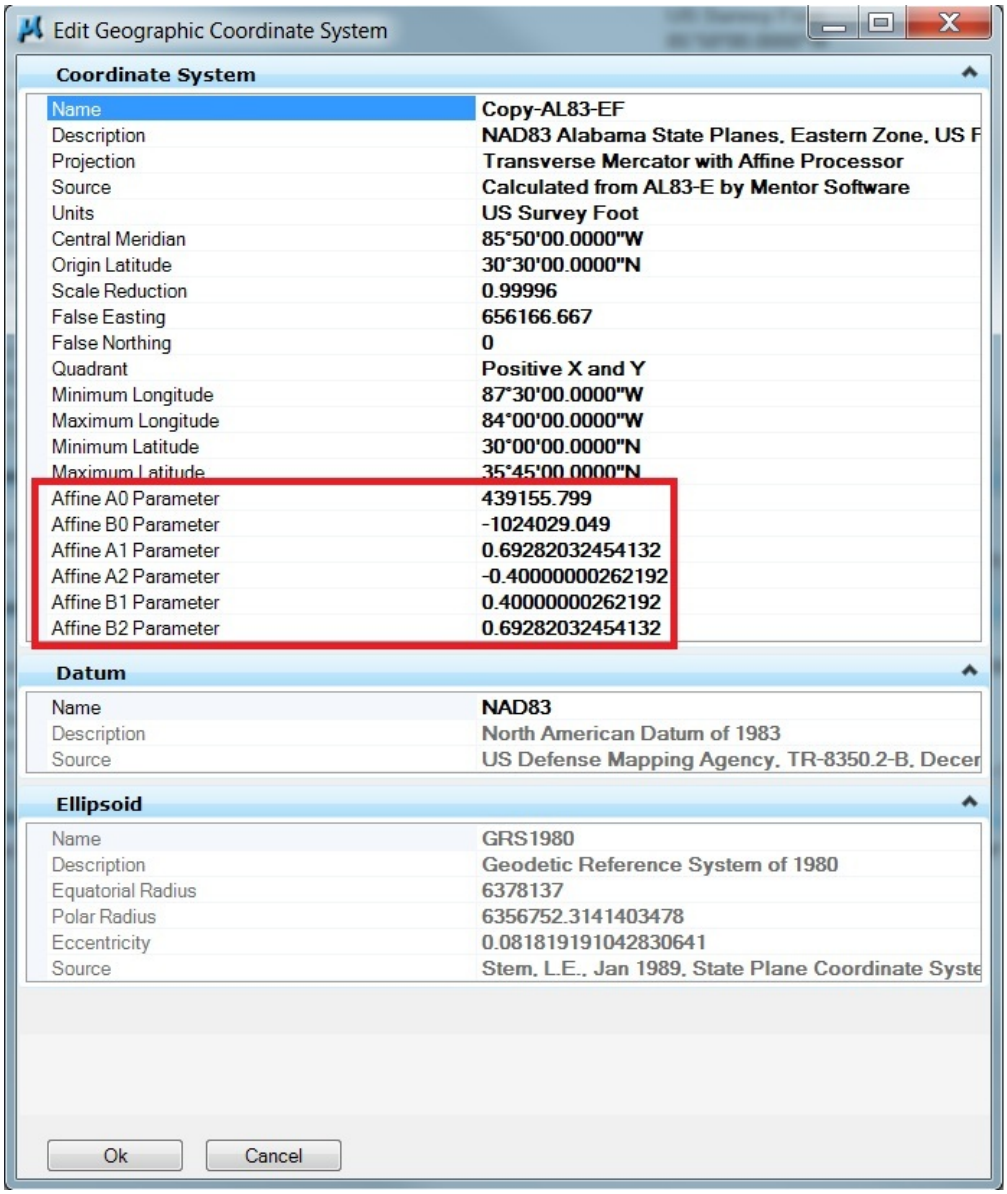


Figure 5

A reprojection can now take place between the custom project coordinate system and any other standard or custom coordinate system. The MicroStation Google Earth tools will also work now for the project coordinate system. However the custom coordinate system cannot be exported to ESRI SHP files using Bentley Map due to the inability to create a PRJ file during the export. The custom coordinate system will have to be reprojected to a standard coordinate system by simply changing the coordinate system and reprojecting before exporting SHP files. The custom coordinate system is also not a valid coordinate system in a georeferenced PDF file.

Using the Helmert Transformation

The 2D Helmert Transformation in MicroStation is the same as the 2D Conformal Transformation with the added capability of modifying elevations by a constant Z offset during the transformation. This added functionality begins with MicroStation V8i Select Series 3, but is not backwardly compatible in Select Series 2 or earlier.

2D Helmert Transformation Equations

$$E = a\,X - b\,Y + t_x$$

$$N = a\,Y + b\,X + t_y$$

$$H = Z + t_z$$

Where:

$a = k \cos \theta$

$b = k \sin \theta$

k = the ratio of the distance between 2 points in a standard coordinate system to the distance between the same 2 points in a local coordinate system

θ = rotation about the origin of the local coordinate system to make it parallel with the standard system (counter-clockwise is positive)

t_x = translation in the X direction applied in the local system after the scale and rotation are applied

t_y = translation in the Y direction applied in the local system after the scale and rotation are applied

t_z = the elevation difference between the standard and local coordinate systems

And:

X and Y are the known coordinates of a point in the local coordinate system.

E and N are the unknown easting and northing of a point in a standard coordinate system.

Z = elevation in the local coordinate system

H = elevation in the standard coordinate system

Also note that the transformation is from the local coordinate system to the standard coordinate system.

After setting the standard coordinate system, select *Details* from the Geographic Coordinate System dialog to add the Helmert Transform.

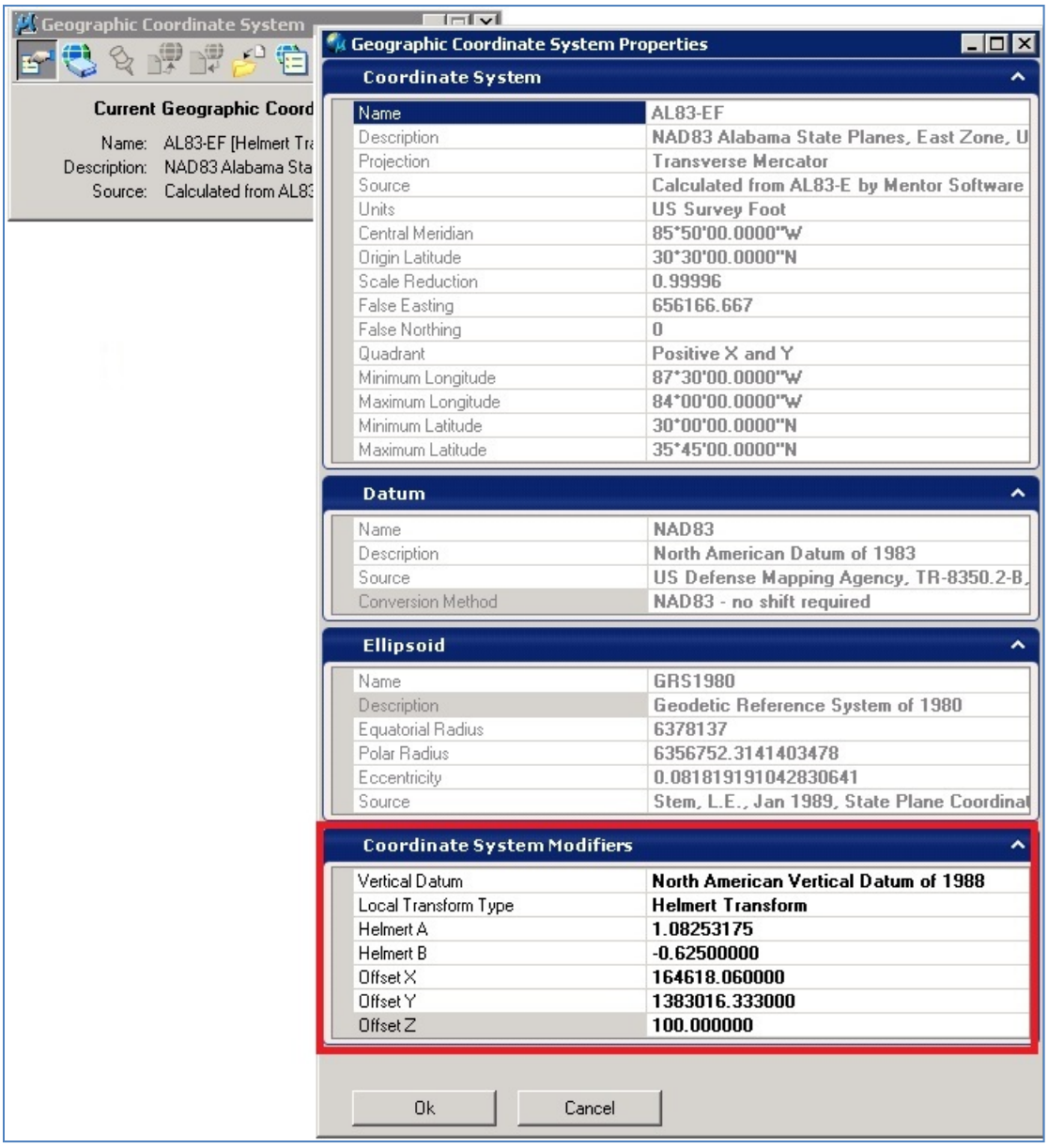


Figure 6

Helmert A = “a” = k Cos Θ

Helmert B = “b” = k Sin Θ

Offset X = “t_x”

Offset Y = "t_y"

Offset Z = the difference in elevation between the standard and local coordinate systems

The same limitations of creating a PRJ file, when exporting to ESRI SHP files with Bentley Map and georeferenced PDF files, exist when using the Helmert Transform as when using Affine Parameters.

Compute the Grid to Ground Scale Factor as an Affine Parameter

In previous discussions and examples the scale was determined indirectly from the coordinates of two pairs of points in a local and standard coordinate system. A direct method will now be used to compute the scale factor.

Previously it was stated that distances of a perfect survey will not match distances on a perfect map using a standard coordinate system. Distances on a map are deliberately scaled in a two stage process. The first stage of the process is to compute a scale factor to project points and lines to an ellipsoid that is used as a mathematical model of mean sea level, Elevation 0.0. For the U.S. the GRS80 ellipsoid of the NAD83 datum or the WGS84 ellipsoid of the WGS84 datum are commonly used.

For a project coordinate system, a mean height above or below the ellipsoid can be used for the entire region. A simple equation can be used to compute the scale factor k_e.

$$k_e = \frac{R}{R + h}$$

Or

$$k_e = \frac{R}{R + H + N}$$

Where:

R = the mean earth radius (6,372,160 m or 20,906,000 ft)

h = ellipsoid height (the height reported by some GPS equipment)

H = elevation (It can be determined from a differential level survey or trigonometric leveling from a benchmark, from some GPS equipment or from photogrammetric mapping. It is also called the *orthometric* height.)

N = geoidal separation of the geoid from the ellipsoid (Separation below the ellipsoid is negative. The separation in the U.S. can be computed from tools at http://www.ngs.noaa.gov/TOOLS/.)

Some GPS equipment report the ellipsoid height while others report the elevation. If the elevation is reported, the geoidal separation N must be determined. Benchmarks are heights above the geoid, not the ellipsoid.

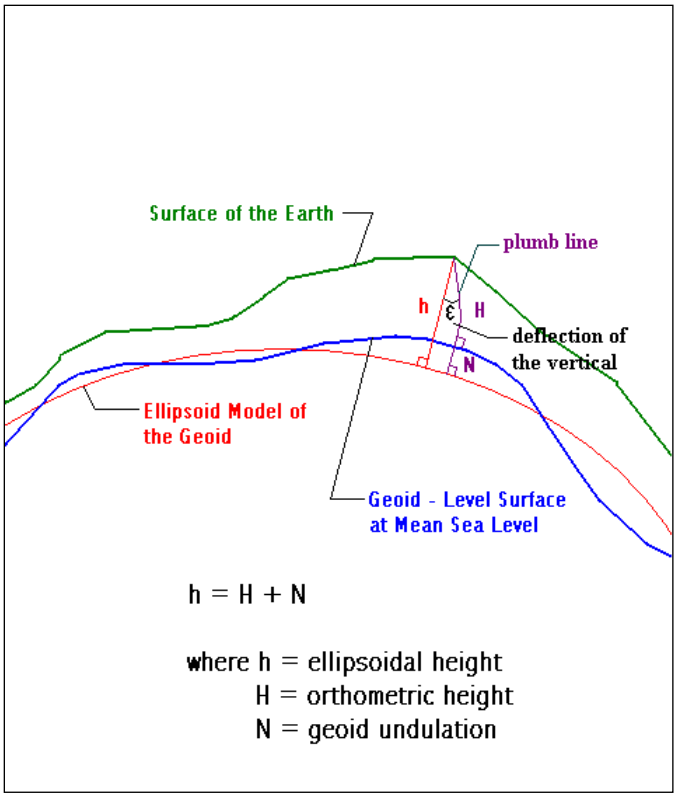


Figure 7

The second stage of the transfer of ground points to map points is the computation of a mean *map scale factor* k_m for the region. The map scale factor is the ratio of distances on the ellipsoid to projected distances on the map. The Transverse Mercator projection and the Lambert Conformal Conic projection are said to be *conformal* because the scale factor is the same in every direction from a point. Computer software is required to compute the map scale factor. "Corpscon" is a free software package, available from the World Wide Web, which will compute the map scale factor and *convergence* at a point. Bentley Map will also compute the map scale and convergence at a point. The convergence is not used in the computation of the Affine Parameters.

Compute the *ground to grid scale factor* for a region by multiplying the two scale factors:

$$k_g = k_e\, k_m$$

Compute the *grid to ground scale factor*:

$$K_G = 1/k_g$$

Affine Parameters for a project coordinate system with no rotation or translation:

$$A0 = 0$$

$$B0 = 0$$

$$A1 = K_G \operatorname{Cos} 0° = K_G$$

$$A2 = -K_G \operatorname{Sin} 0° = 0$$

$$B1 = K_G \operatorname{Sin} 0° = 0$$

$$B2 = K_G \operatorname{Cos} 0° = K_G$$

It may be desirable to translate (Fence Move) the local coordinate system graphics by a large arbitrary amount, such as moving the graphics by a million, in the X and/or Y direction to ensure that coordinates for points in the project coordinate system are not confused with coordinates in the standard coordinate system. In that case, values should be entered for A0 and/or B0 for the X and Y translation.

Handling Redundancy with Least Squares

In the real world after performing a 2D conformal transformation, it may be observed that the two pairs of control points selected to compute the transformation match perfectly, but other pairs of points are shifted slightly from each other. This is due to small but acceptable *random errors* as a result of the mapping techniques, for example, digitizing, surveying, photogrammetry, etc., used to derive the data. The solution is to add pairs of control points to the computation of the transformation. It is best to select pairs of points that uniformly cover the area of interest to be transformed.

A *least squares* solution will compute the transformation parameters that best apply to all coordinate pairs. All pairs of points will likely have *residual* easting and northing between the pairs of points, but the mismatch will be minimized overall.

For three pairs of points:

$$\mathbf{E} = \mathbf{X}\,\boldsymbol{T}$$

$$\mathbf{E} = \begin{vmatrix} E_1 \\ N_1 \\ E_2 \\ N_2 \\ E_3 \\ N_3 \end{vmatrix} \qquad \mathbf{X} = \begin{vmatrix} X_1 & -Y_1 & 1 & 0 \\ Y_1 & X_1 & 0 & 1 \\ X_2 & -Y_2 & 1 & 0 \\ Y_2 & X_2 & 0 & 1 \\ X_3 & -Y_3 & 1 & 0 \\ Y_3 & X_3 & 0 & 1 \end{vmatrix} \qquad \boldsymbol{T} = \begin{vmatrix} a \\ b \\ t_x \\ t_y \end{vmatrix}$$

Note that the $\mathbf{X}$ matrix is no longer square. There are 6 rows and 4 columns. Therefore the matrix cannot be simply inverted to move it to the other side of the equation.

The matrix equation with the row-column dimensions as subscripts:

$${}_6\mathbf{X}_4\,{}_4\boldsymbol{T}_1 = {}_6\mathbf{E}_1$$

Multiply both sides of the equation by $\mathbf{X}^T$:

$${}_4\mathbf{X}^T{}_6\,{}_6\mathbf{X}_4\,{}_4\boldsymbol{T}_1 = {}_4\mathbf{X}^T{}_6\,{}_6\mathbf{E}_1$$

This produces a 4 x 4 matrix ${}_4(\mathbf{X}^T\mathbf{X})_4$ on the left side of the equation which can be inverted.

Multiply both sides of the equation by the inverse $(\mathbf{X}^T\mathbf{X})^{-1}$:

$$(\mathbf{X}^T\mathbf{X})^{-1}\,(\mathbf{X}^T\mathbf{X})\,\boldsymbol{T} = (\mathbf{X}^T\mathbf{X})^{-1}\,\mathbf{X}^T\,\mathbf{E}$$

$(\mathbf{X}^T\mathbf{X})^{-1}\,(\mathbf{X}^T\mathbf{X})$ is the *identity matrix* $\mathbf{I}$ containing all ones on the diagonal with all other matrix elements being zero.

Therefore to solve for $\boldsymbol{T}$ – the transformation parameters:

$$\boldsymbol{T} = (\mathbf{X}^{\mathrm{T}}\,\mathbf{X})^{-1}\,\mathbf{X}^{\mathrm{T}}\,\mathbf{E}$$

This is the matrix equation for an *unweighted least squares adjustment.*

With row-column dimensions for three control point pairs:

$${}_4\boldsymbol{T}_1 = {}_4[({}_4\mathbf{X}^{\mathrm{T}}{}_6\,{}_6\mathbf{X}_4)^{-1}]_4\,{}_4\mathbf{X}^{\mathrm{T}}{}_6\,{}_6\mathbf{E}_1$$

The six equations have been reduced to four equations with four unknowns: "*a*", "*b*", "t_x" and "t_y." For four pairs of points the 6 subscript would change to 8. For five pairs of points the 6 would change to 10. All other matrix dimensions would remain the same. In other words, two equations are available for each pair of points but the overall number will be reduced to four equations with four unknowns.

To transform the XY coordinates of the three local control points, multiply $\mathbf{X}$ and $\mathbf{T}$:

$$\boldsymbol{A} = \mathbf{X}\,\mathbf{T}$$

$${}_6\boldsymbol{A}_1 = {}_6\mathbf{X}_4\,{}_4\mathbf{T}_1$$

For perfect data, matrix $\mathbf{A}$ transformed control points would equal matrix $\mathbf{E}$ control points. Because of random errors in the methods used to determine the coordinates of control points in the local coordinate system and in the standard coordinate system, the least squares solution of the transformation parameters is the statistically "best" solution of the four parameters of $\mathbf{T}$ using the three coordinate pairs. Upon subtracting matrix $\mathbf{E}$ from matrix $\mathbf{A}$, the residual in easting and northing can be determined.

To solve for $\boldsymbol{R}$ – the easting and northing residual of the transformed local points to the standard points:

$$\boldsymbol{R} = \mathbf{A} - \mathbf{E}$$

$${}_6\boldsymbol{R}_1 = {}_6\mathbf{A}_1 - {}_6\mathbf{E}_1$$

A large residual relative to other residuals will be an indication of an error in the data for a pair of points.

$$\boldsymbol{R} = \begin{vmatrix} \Delta E_1 \\ \Delta N_1 \\ \Delta E_2 \\ \Delta N_2 \\ \Delta E_3 \\ \Delta N_3 \end{vmatrix}$$

Statistical Analysis of Least Squares Adjustments

When only two pairs of points are used, the transformed local coordinates will exactly match the standard coordinates, ignoring rounding errors. With least squares, possibly none of the point pairs will exactly match each other. Therefore it is important to determine if the "best" fit is good enough, or perhaps accidental blunders have been introduced or the data are not acceptable. The least squares solution technique and matrix algebra are ideally suited for analyzing results statistically.

The *variance* statistic can be computed when there is redundancy.

Compute the variance of unit weight – the sum of the squares of the residuals divided by the redundancy:

$$S_o^2 = (\mathbf{R}^T\mathbf{R}) \,/\, r$$

Where:

"r" is the redundancy – the number of equations minus the number of unknowns. In the case of 3 pairs of points with 6 equations and 4 unknown transformation parameters, the redundancy is 2 = (6-4).

The *covariance matrix* for the transformation parameters when converting from local coordinates to standard coordinates is:

$$C_T = S_o^2\,(\mathbf{X}^T\mathbf{X})^{-1}$$

$\mathbf{X}$ is the coefficient matrix of the equations used to compute the transformation parameters.

When converting from standard coordinates to local coordinates:

$$C_T = S_o^2\,(\mathbf{E}^T\mathbf{E})^{-1}$$

$\mathbf{E}$ is the coefficient matrix of the equations to compute the transformation parameters.

The covariance matrix:

$$\mathbf{C}_T = \begin{array}{|c|c|c|c|} \hline \mathbf{S_a^2} & S_{a\text{-}b} & S_{a\text{-}tx} & S_{a\text{-}ty} \\ \hline S_{b\text{-}a} & \mathbf{S_b^2} & S_{b\text{-}tx} & S_{b\text{-}ty} \\ \hline S_{tx\text{-}a} & S_{tx\text{-}b} & \mathbf{S_{tx}^2} & S_{tx\text{-}ty} \\ \hline S_{ty\text{-}a} & S_{ty\text{-}b} & S_{ty\text{-}tx} & \mathbf{S_{ty}^2} \\ \hline \end{array}$$

The variances of the 4 transformation parameters are on the diagonal of the covariance matrix. The standard deviations of the transformation parameters can be determined by taking the square root of the values on the diagonal of the covariance matrix. Upon examining the covariance matrix when using actual numbers, it will be observed that $S_a^2 = S_b^2$ and $S_{tx}^2 = S_{ty}^2$ for any covariance matrix of a 2D conformal transformation.

S_a = standard deviation of "a"

S_b = standard deviation of "b"

S_{tx} = standard deviation of "t_X"

S_{ty} = standard deviation of "t_Y"

To compute the standard deviation of the rotation Θ:

$$\Theta = \text{Tan}^{-1} [b / a]$$

From the error propagation of "b" and "a" in the computation of the variance S_θ^2:

$$S_\theta^2 = [a^2 S_a^2 + b^2 S_b^2] / [a^2 + b^2]^2 \text{ (radians)}^2$$

"a" and "b" are from the transformation matrix **T** and S_a^2 and S_b^2 are from the diagonal of the covariance matrix $\mathbf{C}_T$.

Because $S_a^2 = S_b^2$ the equation can be simplified:

$$S_\theta^2 = S_a^2 / (a^2 + b^2) \text{ (radians)}^2$$

$$S_\theta = \sqrt{S_\theta^2} \text{ radians}$$

or

$$S_\theta = 180/\pi \sqrt{S_\theta^2} \text{ degrees}$$

or

$$S_\theta = 10800/\pi \sqrt{S_\theta^2} \text{ minutes}$$

or

$$S_\theta = 648000/\pi \sqrt{S_\theta^2} \text{ seconds}$$

To determine the standard deviation of "k":

$$k = \sqrt{a^2 + b^2}$$

The variance of "k" as a result of error propagation from "a" and "b":

$$S_k^2 = [a^2 S_a^2 + b^2 S_b^2] / [a^2 + b^2]$$

Because $S_a^2 = S_b^2$:

$$S_k^2 = S_a^2 = S_b^2$$

The scale standard deviation:

$$S_k = \sqrt{S_a^2}$$

"a" and "b" are from the transformation matrix **T.** "S_a^2" is from the covariance matrix $\mathbf{C}_T$**.**

Compute the standard deviation of the transformed coordinates:

$$E = a\,X - b\,Y + t_x$$

$$N = b\,X + a\,Y + t_y$$

The uncertainty in the four transformation parameters (that is the standard deviation or variance) results in an uncertainty in the adjusted coordinates. Using the mathematical technique of "*propagation of random errors*," the standard deviations of a point location parallel to the easting and northing axes can be computed.

To compute the standard deviation of the easting and northing of the transformed points:

$$S_E = \sqrt{X^2\,S_a^2 + Y^2\,S_b^2 + S_{tx}^2}$$

$$S_N = \sqrt{X^2\,S_b^2 + Y^2\,S_a^2 + S_{ty}^2}$$

S_a^2, S_b^2, S_{tx}^2, and S_{ty}^2 are from the diagonal of the covariance matrix $\mathbf{C}_T$.

Because $S_a^2 = S_b^2$ and $S_{tx}^2 = S_{ty}^2$:

$$S_E = S_N = \sqrt{S_a^2(X^2 + Y^2) + S_{tx}^2}$$

To verify calculations of the standard deviations S_θ, S_k, and S_{tx}, the standard deviation of the transformed point can be checked using this alternate method:

$$E = (k\,\mathrm{Cos}\,\theta)\,X - (k\,\mathrm{Sin}\,\theta)\,Y + t_x$$

$$S_E^2 = S_N^2 = (X\,\mathrm{Cos}\,\theta - Y\,\mathrm{Sin}\,\theta)^2\,S_k^2 + k^2\,(X\,\mathrm{Sin}\,\theta + Y\,\mathrm{Cos}\,\theta)^2\,S_\theta^2 + S_{tx}^2$$

$$S_E = S_N = \sqrt{S_E^2}$$

Use Matrix Algebra to Calculate Point Variances

The propagation of the uncertainty (variance) from the computed transformation parameters to the transformed point locations can be computed using matrix algebra.

When transforming from a local coordinate system to a standard coordinate system:

$$\mathbf{S_E}^2 = \mathbf{X}\,\acute{\mathbf{C}}_T\,\mathbf{X}^T$$

Where the computed variances of the eastings and northings are on the diagonal of the $\mathbf{S_E}^2$ matrix and the $\acute{\mathbf{C}}_T$ matrix elements that are off the diagonal have been set to zero – that is the transformation parameters are not correlated.

$$\mathbf{X} = \begin{vmatrix} X_1 & -Y_1 & 1 & 0 \\ Y_1 & X_1 & 0 & 1 \\ X_2 & -Y_2 & 1 & 0 \\ Y_2 & X_2 & 0 & 1 \\ . & . & . & . \\ . & . & . & . \\ X_n & -Y_n & 1 & 0 \\ Y_n & X_n & 0 & 1 \end{vmatrix} \qquad \acute{\mathbf{C}}_T = \begin{vmatrix} S_a^2 & 0 & 0 & 0 \\ 0 & S_b^2 & 0 & 0 \\ 0 & 0 & S_{tx}^2 & 0 \\ 0 & 0 & 0 & S_{ty}^2 \end{vmatrix}$$

$$\mathbf{S_E}^2 = \begin{vmatrix} S_{E1}^2 & & & & & \\ & S_{N1}^2 & & & & \\ & & S_{E2}^2 & & & \\ & & & S_{N2}^2 & & \\ & & & & \ddots & \\ & & & & & S_{En}^2 \\ & & & & & & S_{Nn}^2 \end{vmatrix}$$

When transforming from a standard coordinate system to a local coordinate system:

$$\mathbf{S_X}^2 = \mathbf{E}\,\acute{\mathbf{C}}_T\,\mathbf{E}^T$$

Example 4 – Compute Affine Parameters for a Project Coordinate System using Least Squares

Compute the Affine Transformation Parameters

Using the same data from Example 3, add the coordinate pair of the lower right corner of the square as an additional control point to the **E** coefficient matrix and its corresponding control point to the **X** column matrix and perform a transformation from the standard coordinate system **EN** to the local project coordinate system **XY**. See Figure 2. Random errors have been simulated in the **X** column matrix (local project coordinate system).

To compute the transformation parameters from the standard coordinate system to the local project coordinate system:

$$\boldsymbol{T} = (\mathbf{E}^{\mathrm{T}}\mathbf{E})^{-1}\mathbf{E}^{\mathrm{T}}\mathbf{X}$$

$$\mathbf{X} = \begin{bmatrix} 100000.25 \\ 199999.75 \\ 103999.75 \\ 204000.25 \\ 104000.25 \\ 200000.00 \end{bmatrix} \qquad \mathbf{E} = \begin{bmatrix} 397871.235 & -1537022.683 & 1 & 0 \\ 1537022.683 & 397871.235 & 0 & 1 \\ 404701.362 & -1538852.810 & 1 & 0 \\ 1538852.810 & 404701.362 & 0 & 1 \\ 402201.362 & -1534522.683 & 1 & 0 \\ 1534522.683 & 402201.362 & 0 & 1 \end{bmatrix}$$

$$\boldsymbol{T} = \begin{bmatrix} 0.69278739984777 \\ 0.40008202984292 \\ 439295.164 \\ -1024011.392 \end{bmatrix}$$

$\theta = \text{Tan}^{-1}$ [b / a]

$\theta = 30.0062667366° = 30°\ 00'\ 22.6"$ counter-clockwise

$k = \text{a} / \text{Cos}\ \theta$ or $k = \text{b} / \text{Sin}\ \theta$ or $k = \sqrt{a^2 + b^2}$

Scale = 0.800012507

Affine Parameters for a project coordinate system from the T matrix:

A0 = 439295.164

B0 = −1024011.392

A1 = 0.69278739984777

A2 = −0.40008202984292

B1 = 0.40008202984292

B2 = 0.69278739984777

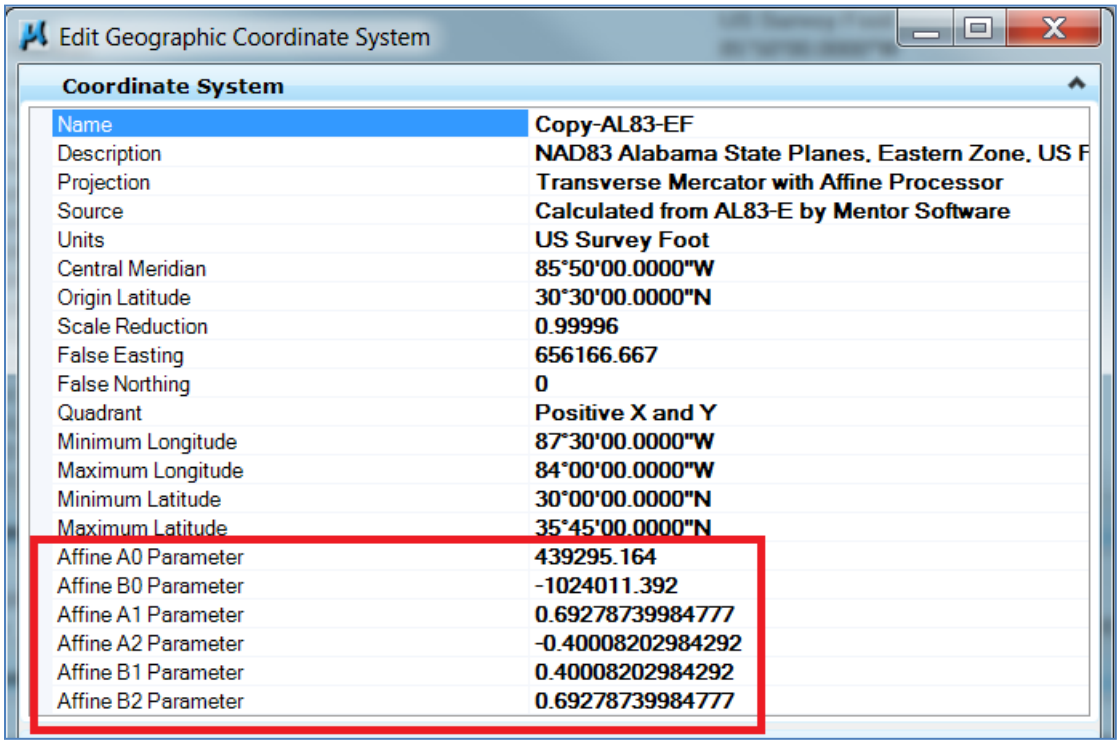


Figure 8

Compute the Standard Deviations for the Transformation Parameters

Compute the easting and northing residuals for the three pairs of points.

$$\boldsymbol{R} = \mathbf{E}\,\mathbf{T} - \mathbf{X}$$

Compute the transformed points **ET** in the local coordinate system and subtract the local coordinates prior to the adjustment to get the residuals:

$$\mathbf{ET} = \begin{bmatrix} 100000.1875 \\ 199999.6875 \\ 103999.8125 \\ 204000.1875 \\ 104000.2500 \\ 200000.1250 \end{bmatrix} \qquad \mathbf{X} = \begin{bmatrix} 100000.25 \\ 199999.75 \\ 103999.75 \\ 204000.25 \\ 104000.25 \\ 200000.00 \end{bmatrix}$$

$$\boldsymbol{R} = \begin{bmatrix} -0.0625 \\ -0.0625 \\ 0.0625 \\ -0.0625 \\ 0.0000 \\ 0.1250 \end{bmatrix}$$

Compute the variance of unit weight:

$$S_o^2 = (\mathbf{R}^T\mathbf{R}) \,/\, r$$

$$S_o^2 = 0.03125/(6-4)$$

$$S_o^2 = 0.015625$$

$$S_o = \pm 0.125 \text{ ft}$$

Compute the covariance matrix $\boldsymbol{C}_T$:

$$\boldsymbol{C}_T = S_o^2\,(\mathbf{E}^T\mathbf{E})^{-1}$$

$$\boldsymbol{C}_T = \begin{bmatrix} 0.0000000004687500 & 0 & -0.0001882459335940 & -0.0007203747245821 \\ 0 & 0.0000000004687500 & 0.0007203747245927 & -0.0001882459335977 \\ -0.0001882459336001 & 0.0007203747245911 & 1182.675 & 0 \\ -0.0007203747245805 & -0.0001882459336038 & 0 & 1182.675 \end{bmatrix}$$

$S_a^2 = S_b^2 = 0.00000000046875$

$S_{tE}^2 = S_{tN}^2 = 1182.675 \text{ ft}^2$

Compute the standard deviation of the east and north translation:

$S_{tE} = S_{tN} = \sqrt{1182.675} = \pm 34.390$ ft

Compute the standard deviation of the rotation:

$$S_{\theta^2} = S_a^2 \,/\, (a^2 + b^2)\ (\text{radians})^2$$

$$S_{\theta^2} = \frac{0.00000000046875}{(0.69278739984777)^2 + (0.40008202984292)^2}$$

$$S_{\theta^2} = 0.000000000732398974\ (\text{radians})^2$$

$$S_{\theta} = \pm 0.0000270628708 \text{ radians}$$

$$S_{\theta} = \pm\ (648000/\pi)\ (0.0000270628708) \text{ seconds} = \pm 5.6"$$

Compute the standard deviation of the scale:

$$S_{k^2} = S_a^2$$

$$S_{k^2} = 0.00000000046875$$

$$S_k = \pm 0.000021651$$

Compute the Standard Deviations for the Transformed Points

From the error propagation of the transformation parameters, compute the standard deviation of the transformed point locations:

$$S_X = S_Y = \sqrt{S_a^2 (E^2 + N^2) + S_{tE}^2}$$

For Point 1:

$S_{X1} = \sqrt{S_a^2 (E_1^2 + N_1^2) + S_{tE}^2}$

$S_{X1} = [(0.0000000004687500) [(397871.235)^2 + (1537022.683)^2] + 1182.675]^{1/2}$

$S_{X1} = S_{Y1} = \pm 48.624$ ft

For Point 2:

$S_{X2} = \sqrt{S_a^2 (E_2^2 + N_2^2) + S_{tE}^2}$

$S_{X2} = [(0.0000000004687500) [(404701.362)^2 + (1538852.810)^2] + 1182.675]^{1/2}$

$S_{X2} = S_{Y2} = \pm 48.677$ ft

For Point 3:

$S_{X3} = \sqrt{S_a^2 (E_3^2 + N_3^2) + S_{tE}^2}$

$S_{X3} = [(0.0000000004687500) [(402201.362)^2 + (1534522.683)^2] + 1182.675]^{1/2}$

$S_{X3} = S_{Y3} = \pm 48.603$ ft

Compute the Standard Deviations for the Transformed Points Using Matrix Algebra

$$\mathbf{S_X^2 = E\,\acute{C}_T\,E^T}$$

$$\mathbf{E} = \begin{bmatrix} 397871.235 & -1537022.683 & 1 & 0 \\ 1537022.683 & 397871.235 & 0 & 1 \\ 404701.362 & -1538852.810 & 1 & 0 \\ 1538852.810 & 404701.362 & 0 & 1 \\ 402201.362 & -1534522.683 & 1 & 0 \\ 1534522.683 & 402201.362 & 0 & 1 \end{bmatrix}$$

$$\mathbf{\acute{C}_T} = \begin{bmatrix} 0.0000000004687500 & 0 & 0 & 0 \\ 0 & 0.0000000004687500 & 0 & 0 \\ 0 & 0 & 1182.675 & 0 \\ 0 & 0 & 0 & 1182.675 \end{bmatrix}$$

$$\mathbf{S_X^2} = \begin{bmatrix} 2364.2716 & 0.0000 & 2366.8640 & -4.5796 & 2363.2780 & -3.5860 \\ 0.0000 & 2364.2716 & 4.5796 & 2366.8640 & 3.5860 & 2363.2780 \\ 2366.8640 & 4.5796 & 2369.4798 & 0.0000 & 2365.8821 & 0.9819 \\ -4.5796 & 2366.8640 & 0.0000 & 2369.4798 & -0.9819 & 2365.8821 \\ 2363.2780 & 3.5860 & 2365.8821 & -0.9819 & 2362.2961 & 0.0000 \\ -3.5860 & 2363.2780 & 0.9819 & 2365.8821 & 0.0000 & 2362.2961 \end{bmatrix}$$

$S_{X1} = S_{Y1} = \sqrt{2364.2716} = \pm\, 48.624$ ft

$S_{X2} = S_{Y2} = \sqrt{2369.4798} = \pm\, 48.677$ ft

$S_{X3} = S_{Y3} = \sqrt{2362.2961} = \pm\, 48.603$ ft

The results agree with the previous manual calculations.

Summary of Transformation Parameter Results

Transformation Parameter	Computed Value	Confidence Interval			
		68 %	90 %	95 %	99 %
		S	1.645 S	1.960 S	2.576 S
θ	30° 00' 22.6"	± 5.6"	± 9.2"	± 11.0"	± 14.4"
k	0.800012507	± 0.000021651	± 0.000035616	± 0.000042436	± 0.000055773
t_E	439295.164 ft	± 34.390 ft	± 56.572 ft	± 67.404 ft	± 88.589 ft
t_N	-1024011.392 ft	± 34.390 ft	± 56.572 ft	± 67.404 ft	± 88.589 ft

Summary of Control Point Transformation Results

Point	Transformed Value (ft)	Residual (ft)	Confidence Interval			
			68 %	90 %	95 %	99 %
			S (ft)	1.645 S (ft)	1.960 S (ft)	2.576 S (ft)
X_1	100000.1875	-0.0625	± 48.624	± 79.986	± 95.303	± 125.255
Y_1	199999.6875	-0.0625	± 48.624	± 79.986	± 95.303	± 125.255
X_2	103999.8125	0.0625	± 48.677	± 80.074	± 95.408	± 125.393
Y_2	204000.1875	-0.0625	± 48.677	± 80.074	± 95.408	± 125.393
X_3	104000.2500	0.0000	± 48.603	± 79.953	± 95.263	± 125.203
Y_3	200000.1250	0.1250	± 48.603	± 79.953	± 95.263	± 125.203

Appendix

Matrix Calculations in Microsoft Excel

Matrix calculations can be performed in the Microsoft Excel spreadsheet software. See the matrix calculations for the examples using Microsoft Excel.

Spreadsheet cell designation

A cell location is specified by the column letter followed by the row number – such as "A23", "K4", etc.

Rectangular Array Designation

A rectangular array is designated by two cell locations: the upper left cell location of the array and the lower right cell location. The two cell locations are separated by a colon, such as "B2:C5." For a column matrix, the column letter would be the same, such as "A3:A8." For a row matrix, the row number would be the same, such as "C3:F3."

Workflow

1. Enter the matrix data into a rectangular *Array* of cells in the spreadsheet. Depending upon the matrix operation being performed, one or two arrays will be required. See Figure 9 below.

2. Drag the cursor with the mouse to select a rectangular array of cells to hold the output matrix. The number of rows and columns of the output matrix must be known in advance.

3. With the output array selected, key in the function for the matrix operation.

Operation	Key In the function	Number of Input Arrays
Add	=*Array1* + *Array2*	2
Subtract	=*Array1* - *Array2*	2
Transpose	+TRANSPOSE (*Array*)	1
Multiply	+MMULT (*Array1* , *Array2*)	2
Invert	+MINVERSE (*Array*)	1
Scale	=*Scale* * *Array*	1

4. Initiate the operation by simultaneously pressing "Ctrl – Shift – Enter." See Figure 10 below.

5. Modify the width of columns of the results if required.

6. Optionally modify the format and/or precision of the data by selecting a cell or cells and right-click with the mouse. Select "Format Cells" from the popup menu and then click "Number" on the *Number* tab in the *Format Cells* dialog.

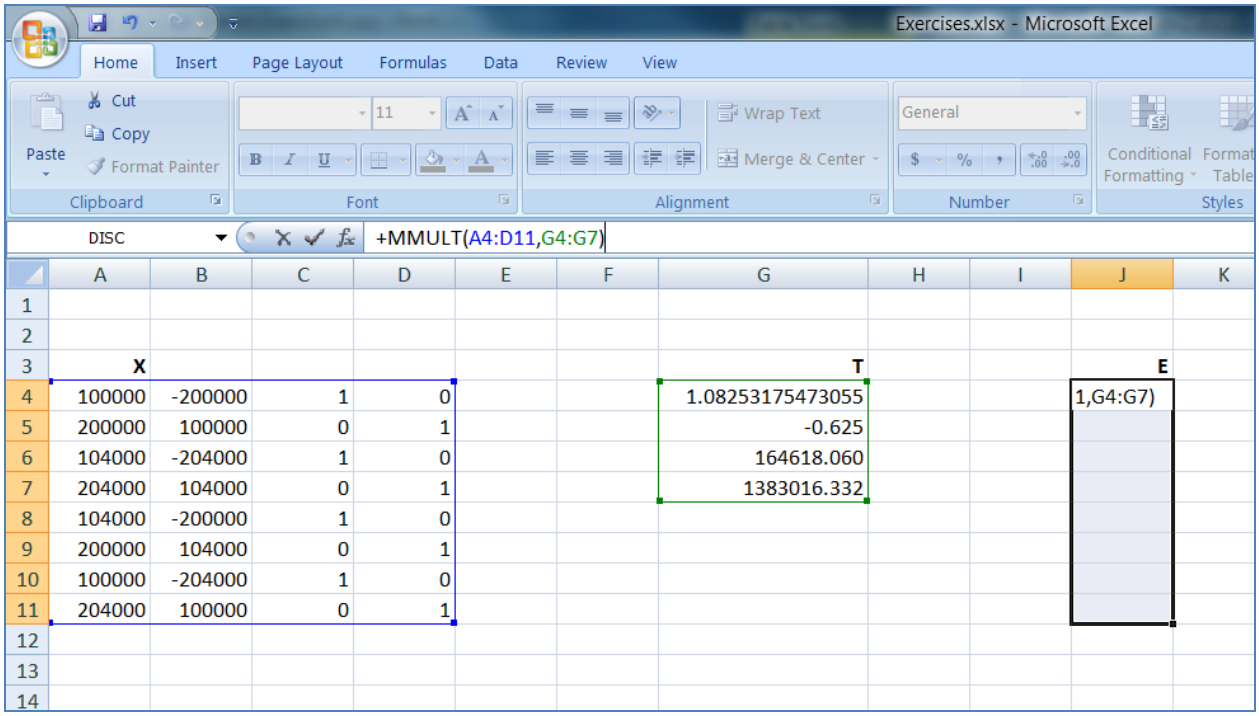


Figure 9

Press “Ctrl – Shift – Enter” to initiate the calculation.

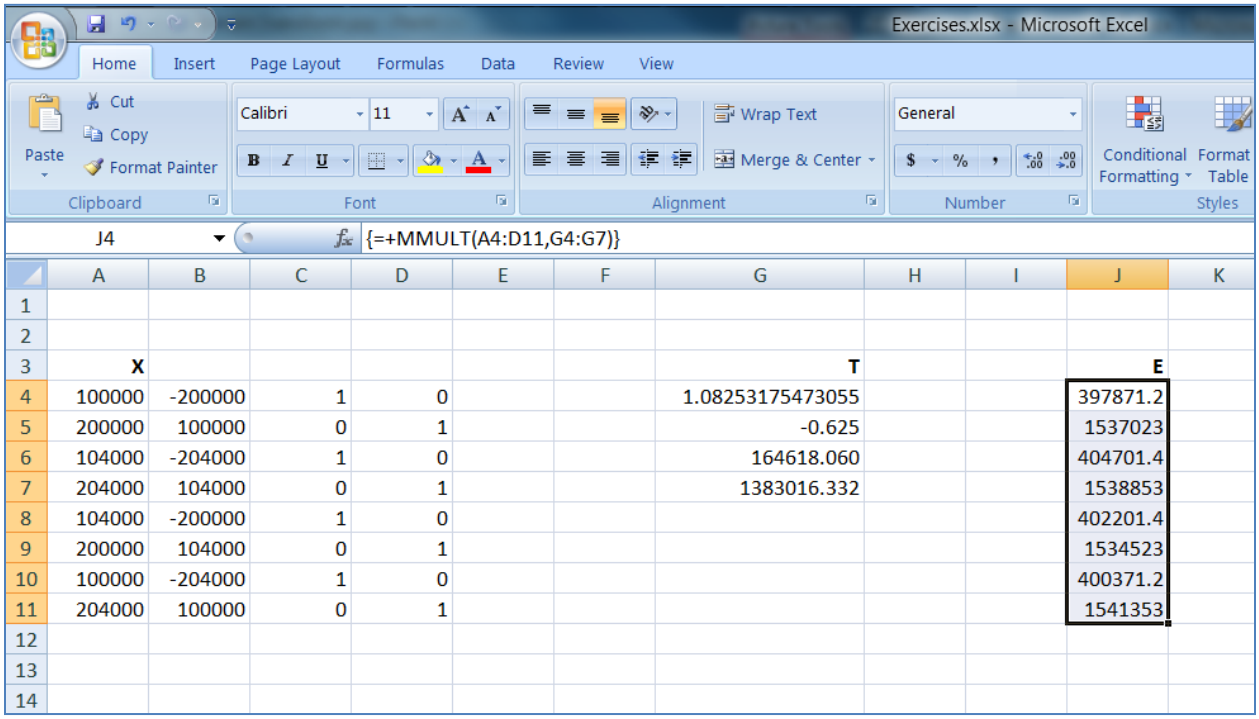


Figure 10

Predetermine the Dimensions of the Output Matrix

In step 2 above the size of the output matrix had to be known in advance. Use the following to determine the output matrix dimensions.

- Addition and Subtraction – All 3 matrices (2 input and 1 output) must have the same row and column dimensions.

- Transpose – The number of rows and columns are reversed from the input matrix to the output matrix.

$$_{a}\mathbf{E}_{b} \qquad _{b}\mathbf{E}^{T}{}_{a}$$

- Multiply – The number of columns in the first array must equal the number of rows in the second array. The dimensions of the output matrix array will be the number of rows of the first array and the number of columns of the second array.

$$_{i}\mathbf{X}_{j}\ _{j}\mathbf{T}_{k} = {}_{i}\mathbf{E}_{k}$$

  When multiplying a row matrix times a column matrix, for example $_{1}\mathbf{R}^{T}{}_{n}\ {}_{n}\mathbf{R}_{1}$, the output will be a single cell.

- Invert – The matrix array must be square – number of columns must equal number of rows. The output matrix will have the same dimensions.

- Scale – The input and output matrix arrays have the same dimensions.